

vizzes for **Visualizations**

Big Data Analytics / Data Science

Presentation Phase

Ready Results; Known Insights; Clean DataSets

Which Viz will you choose and why?

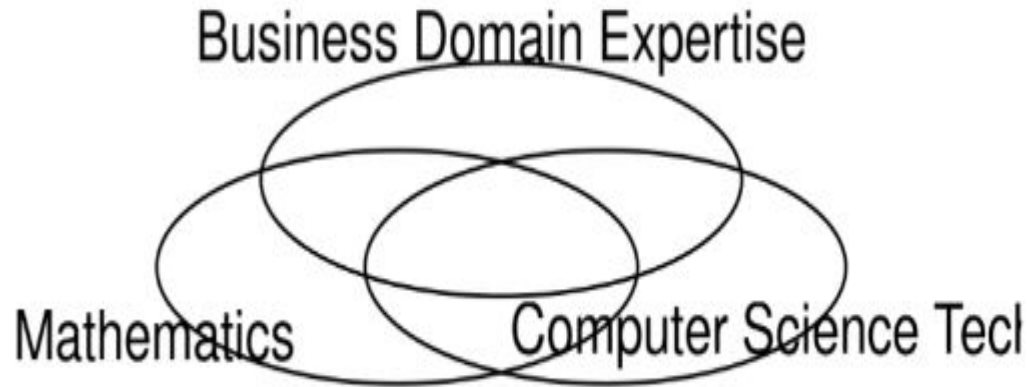
Example 1

The Data Science has below three equally important primary aspects involved.

- Mathematics
- Computer Science Technology
- Subject matter expertise

There are other internal sub aspects like Machine Learning, where NO involvement of human/subject matter expertise but only Mathematics and Computer Science Technology.

Venn diagram (Think what can be improved here)

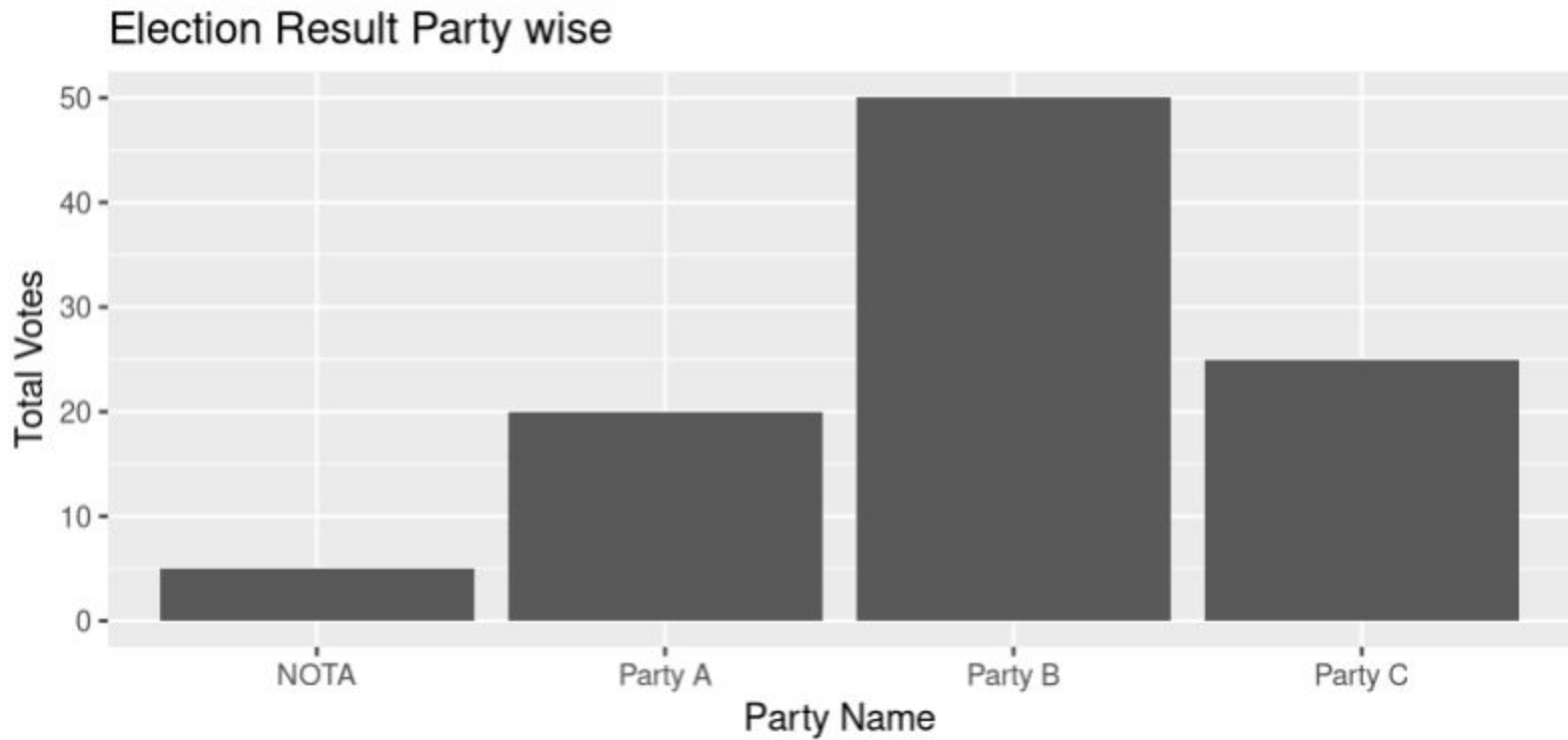


[my-dev/R/VennDiagram](#)

Example 2:

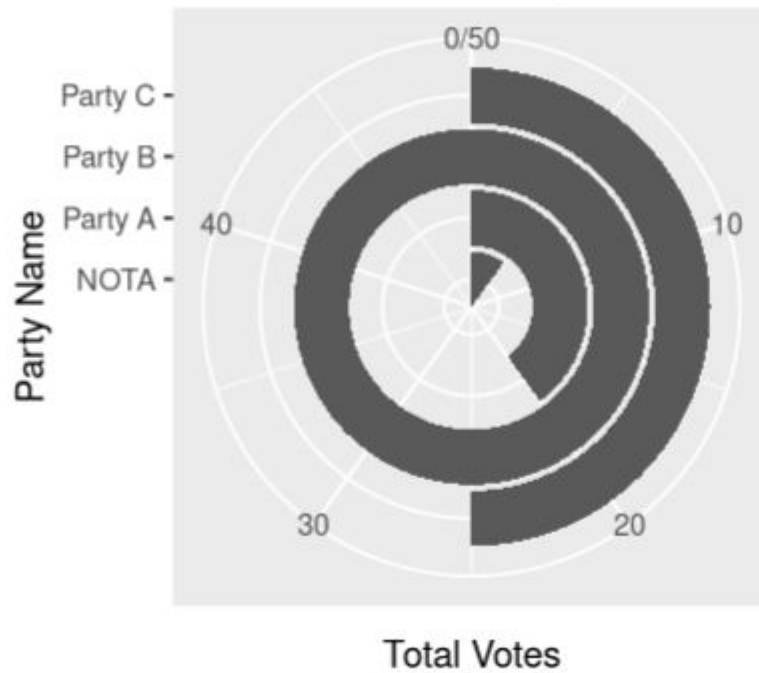
Voting Results

Party Name	Votes
Party A	20
Party B	50
Party C	25
NOTA	5

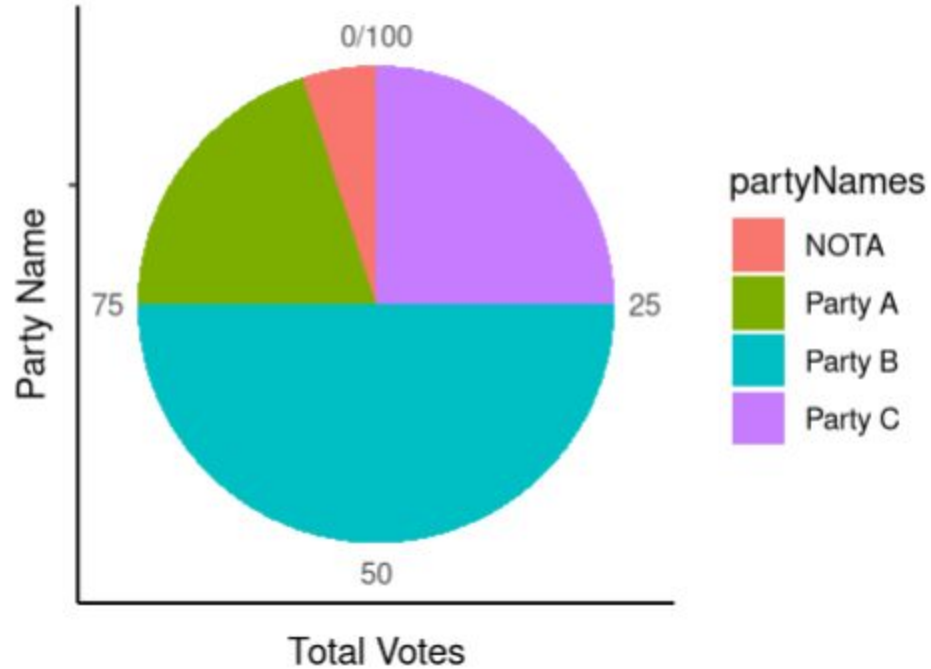


How about some ordering/sorting?

Election Result Party wise



Election Result Party wise

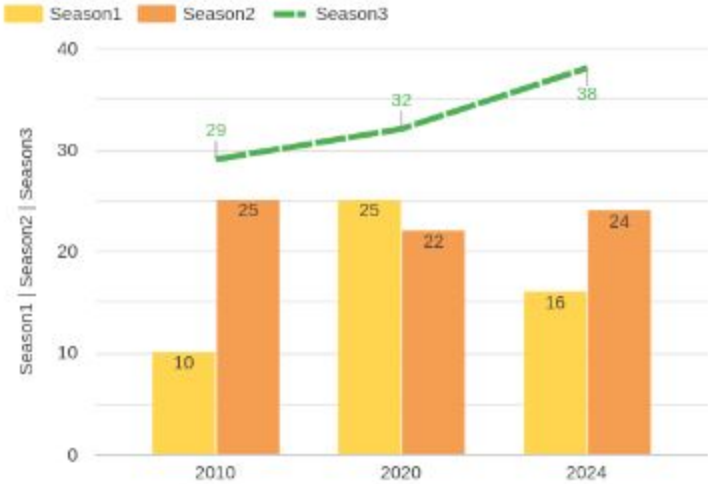


Example 3:

Every year during Winter season, Sales is high and every year it increases too.
Note that sales over years may or may not have increased.

SalesYear	Season1	Season2	Season3
2010	10	25	29
2020	25	22	32
2024	16	24	38

Sales over years per season



Crated using google looker studio web

Example 4:

Known that participants during a technical workshop have practiced many computer commands. But some of them are trending and useful to many participants. Showcase top n most used commands from participants' work history.

WordCloud/TagCloud showcase words with bigger font size having higher frequencies of occurrence



my-dev/R/participantsWorkRTagCloud

Example 5:

CO2 Per Capita needs to be highlighted interactively from 2000 to 2011 years including all if chosen for the world.

Sheet 1



CO2 Per Capita (metric to...



Year of Year

☐ (All)

☐ 2000

☐ 2001

☐ 2002

☐ 2003

☐ 2004

☐ 2005

☐ 2006

☐ 2007

☐ 2008

☐ 2009

☐ 2010

☒ 2011

© 2025 Mapbox © OpenStreetMap

[View on Tableau Public](#)

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Example 6:

University dataset is available and interactive report is expected to show university wise CS/Non-CS students details.

Coffee_Landscape.jrxml

MongoDB Connectio... Java Page 1 of 2C 100%

Input Parameters

Background(CS\NCS)

CS

Reset

Student

University Group

The youth power.

First Name	Middle Name	Last Name	Background(CS\NCS)	TrainingExamAvgPerc
ADU				
Ruben	Gershone	Chaudhari	CS	88
ANU				
Pavithra		Gangadharan	CS	94
Ilampirai		Ganesan	CS	93
Premalatha	Soreka	Ganesh	CS	93
Anil		Dhawan	CS	91
Ranjit	Singh	Dhillon	CS	90
Sindhu		Gangadhar	CS	90
Ranjith		Dubey	CS	89
Dinesh	Babu	Ganesh	CS	89
Satheeshraj		Ganguly	CS	85
Suchithra		Gandhi	CS	84
Leno	Sharon	Fernandes	CS	81
Gokul		Eswaramoorthy	CS	80
Bharathwaj	Radhakrishnan	Gangil	CS	79
Mayur		Dhiman	CS	67
AU				
Ashok	Reddy	Dhall	CS	85

Design Source Preview JasperReports Library

<https://community.jaspersoft.com/wiki/designing-report-jaspersoft-studio>
<https://community.jaspersoft.com/wiki/jaspersoft-mongodb-query-language>

Text Book: Big Data And Analytics by Subhashini Chellappan Seema Acharya

It's been said that “A picture is worth a thousand words”

But for those who are willing to understand and highly gifted.

For others, it's still a challenge.



And this is why even designing visualization is a challenge
and more than just an art.

Warning #1 Never represent non-genuine matter.

Warning #2 WIN-WIN for humanity and world a must.

Warning #3 Beware of misunderstanding (misrepresentation).

#EthicalVisulization

Various aspects for conveying information

- Shape
- Size
- Color
- Orientation/Direction
- Data characteristics (Discrete Vs Continuous, etc)
- Ordering/Reversing
- Grouping
- Reserving
- Border, scale/zoom, shadow, etc.
- locales

General Guidelines

- Have a title, subtitle and legends wherever applicable
- Make sure the baseline for bar chart is appropriate
- Make sure the pie chart accumulates to total 100%
- Provide support to download in various formats
- Provide support to view/access more context/details if any.
- Have input/feedback from group of people/peer blind after seeing draft viz
- Give about 5-10 seconds after presenting visualization and then explain
- Put yourself in shoe of viewers (end user,client,leadership) and viceversa when grasping someone else viz yourself
- Accessibility Support
- Date of last modification and others

Interesting facts

- Google looker has 48 chart types as of Aug'25
- Apache has data exploration and visualization platform called Superset
- Jaspersoft (2011 A.D.) and Tableau are independent product
- There can be grammar of graphics like ggplot2 talks about (i.e. layering)
- Google Visualization API, Motion charts and Fusion charts are grown up
- Integration to primary website/app seems secondary
- Interactive Vizs are simply famous
- Informatica got Built-In Reporting capabilities
- A single company also uses more than one tools in harmony and happily
- Dashboards have to be dashing
- ETL is close to all reporting/visualization tools
- Visuals can help during various phases of analytics and presentation is itself a phase!



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<https://github.com/profjigarpandya/>

kaggle

<https://www.kaggle.com/jigarpandya>

tableau⁺public

<https://public.tableau.com/app/profile/jigarpandya/vizzes>

Tuesday

September 23 2025